

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Pharmaceutical Technology)

1st Semester of M. Pharm (Pharmaceutics(DRA))

University Theory Examination December 2016

PHCC002 Biopharmaceutics & Pharmacokinetics

Date: 14.12.2016 , Wednesday

Time:10:00 a.m to 1:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** Discuss the physicochemical factors affecting drug absorption **10**
- Q.2 A)** What are phase II reaction? Discuss its importance and explain glucuronidation in brief. **05**
- B)** Enumerate factors affecting renal excretion and discuss effect of plasma drug concentration in context to renal excretion. **05**

OR

- Q.2 A)** Write a brief note on the volume of distribution. **05**
- B)** Discuss plasma level time study as method of bioavailability measurement. **05**
- Q.3 Answer the following (Any three)** **15**
- (A)** Write in brief about absorption of drug by Intraocular route of drug administration.
- (B)** In situ methods for estimation of drug absorption
- (C)** Discuss any two techniques to improve bioavailability of drug
- (D)** Write a short note on Latin square design for determination of bioequivalence

SECTION-II

- Q.4** What is IVIVC? Discuss its importance and methods to determine IVIVC **10**
- Q.5** Explain various pharmacokinetic parameters of one compartment open model for IV bolus injection in detail. **10**

OR

- Q.5** Draw a neat diagram showing plasma concentration time profile for a drug given by constant rate IV infusion (one compartment model). Write the formula for loading dose and constant **10**
- Q.6 Answer the following (Any three)** **15**
- (A)** Briefly explain two compartment open and close models
- (B)** Describe Wagner Nelson Method for determination of K_a
- (C)** Discuss reasons for non linear pharmacokinetics. Michaelis menten constant
- (D)** Physiological models for Pharmacokinetics study

